Mohamed Eltahir Makki Elmanna

Lecturer of Biomedical Engineering | PhD Candidate

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Nationality: Sudanese | **Visa Status:** Transferable Work Visa

Driving License: Valid Saudi Driving License

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ResearchGate:	https://www.researchgate.net/profile/Mohamed-Elmanna/

Professional Summary

Ph.D. candidate in Biomedical Engineering specializing in **deep learning, computer vision, and medical image analysis** for digital pathology and hematology. My research focuses on **stain normalization, domain generalization, and graph-based segmentation and classification models** for robust clinical AI systems. I also have experience as a **Lecturer in Biomedical Engineering**, teaching digital image processing and supervising undergraduate research projects. Proficient in **TensorFlow and PyTorch,** with a strong emphasis on rigorous, reproducible research.

Education

Doctor of Philosophy (Ph.D.) in Systems and Biomedical Engineering

Cairo University, Egypt | Oct 2019 – Present
GPA: 3.8/4
Selected Courses: Artificial Intelligence in Biomedical Applications, Optimization, Ultrasonic Tissue Characterization, Advanced Computer-Assisted Diagnosis, Probability & Statistics, Hospital Information Systems.

Master of Science in Systems and Biomedical Engineering

Cairo University, Egypt | Oct 2011 – Jan 2014
Thesis: Computer-Aided Diagnosis System for Digital Mammography
Advisor: Prof. Yasser M. Kadah | GPA: 3.5/4

Bachelor of Science in Biomedical Engineering

Sudan University of Science and Technology, Sudan | Oct 2004 – Oct 2009
Graduation Project: Nurse Calling System | GPA: 3.08/4 (Top of Class)

Work Experience

Lecturer — Biomedical Engineering Department, AlMughtaribeen University, Sudan
Jan 2017 – June 2019

- Delivered lectures and labs in **Digital Image Processing**, **Advanced topics in Biomedical Engineering**, **Hospital Engineering**, and **Quality Management**.
- Supervised **seven undergraduate projects** focused on medical AI analysis and signal processing.
- Mentored projects in **medical image segmentation**, **glaucoma detection**, **leukemia cell segmentation**, **arrhythmia classification**, and **CT-based pulmonary nodule detection**.
- Developed strong skills in **research supervision, technical writing, and scientific communication**.

Biomedical Engineer — Ribat University Hospital, Sudan
Nov 2009 – Nov 2010

- Maintained and calibrated ICU, OR, and sterilization equipment; trained clinicians on proper device operation; ensured biomedical equipment safety and documentation.

Technical Skills

Programming & Frameworks: Python, TensorFlow, PyTorch, Keras, NumPy, OpenCV, Pandas, Matplotlib, MATLAB

Core Research & Applied Skills: Medical Image Analysis • Deep Learning • Computer Vision • CNNs & GANs • Segmentation & Classification • Domain Adaptation • Stain Normalization • Model Evaluation & Optimization • Dataset Curation & Augmentation

Tools & Systems: Linux • Git • Jupyter • Google Colab • Kaggle • Lightning AI

Professional Skills: Scientific Writing • Research Communication • Mentorship • Cross-Disciplinary Collaboration • Problem Solving

Selected Research Projects

- **Graph-Based Hierarchical Multi-Stain CycleGAN (GHMS-CycleGAN):** Designed a novel CycleGAN model integrating hierarchical, graph-based, and auxiliary classification losses for stain normalization and image classification in digital pathology.
  Technology stacks: Python, PyTorch, TensorFlow, OpenCV, Scikit-learn.
- **Deep Learning Segmentation and Classification of RBCs:** Built CNN-based models using a multi-scanner RBC dataset (100K+ cells), achieving domain-generalized performance for diagnostic workflows.
  Technology stacks: Python, TensorFlow, OpenCV, Scikit-learn.
- **Computer-Aided Diagnosis System for Digital Mammography:** Developed CAD system for screening mammography to classify suspicious regions in mammogram to mass or normal regions with an accuracy reach 96%.
  Technology stacks: MATLAB.
- **Medical Image Reconstruction Projects:** Gained strong theoretical and practical implementation skills for medical image reconstruction in CT, MRI, and Ultrasound.
  Technology stacks: MATLAB.

Selected Publications

1. **M. Elmanna**, A. Elsafty, Y. Ahmed, M. Rushdi, A. Morsy, "GHMS-CycleGAN: Graph-Based Hierarchical Multi-stain CycleGAN for Stain Normalization and Classification in Digital Pathology," Journal of Imaging Informatics in Medicine, 2026. https://doi.org/10.1007/s10278-026-01854-x
2. R. Ahmed, A Iltaf, **M. Elmanna**, G. Zhao, H. Li, Y. Du, B. Li, S. Zhou, "MSA-UNet3+: Multi-Scale Attention UNet3+ with New Supervised Prototypical Contrastive Loss for Coronary DSA Image Segmentation," Biomedical Signal Processing and Control, 2026. https://doi.org/10.1016/j.bspc.2026.110539
3. **M. Elmanna**, Y. M. Kadah, "Implementation of Practical Computer-Aided Diagnosis System for Classification of Masses in Digital Mammograms," Proc. ICCNEEE, Khartoum, Sudan, pp. 336–341, 2015. https://doi.org/10.1109/ICCNEEE.2015.7381387